Classification of quantum graphs over complex 3×3 matrices

(Klasyfikacja grafów kwantowych nad
zespolonymi macierzami 3×3)

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Master's thesis

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Abstract

According to all known laws of aviation, there is no way a bee should be able to fly. Its wings are too small to get its fat little body off the ground. The bee, of course, flies anyway because bees don't care what humans think is impossible.

Według wszystkich znanych praw lotnictwa, pszczoła nie powinna w ogóle latać. Jej skrzydła są zbyt małe, by unieść jej pulchne ciałko z ziemi. Pszczoła, oczywiście, lata mimo wszystko, ponieważ pszczoły nie przejmują się tym, co ludzie uważają za niemożliwe.

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Introduction

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Preliminaries

2.1 Group theory

In the following section we will introduce some group theoretic notions that be used throughout this thesis. We assume that the Reader is familiar with the notion of a group.

Definition 2.1.1 (Dual of a group)**.** Let G be a group. The *dual group* $\hat{G}$ is the set of all homomorphisms from G into the multiplicative group of roots of unity in $\mathbb{C}$ with the group operation defined as

$$(\varphi\psi)(g) = \varphi(g)\psi(g)$$

for all $\varphi, \psi \in \hat{G}$ and $g \in G$, where the latter multiplication takes place in $\mathbb{C}$.

Remark 2.1.1. We will often call the elements of the dual group *characters*. If a group is isomorphic to its complement it will be called *self-dual*.

The following proposition will be used throughout the thesis, as almost all of the groups we will be dealing with will be finite and abelian.

Proposition 2.1.1. *If G is finite and abelian, then it is self-dual.*

Proof. First we will prove the statement for finite cyclic groups. Let $G = \langle g \rangle$, $\operatorname{ord}(g) = n$. We will prove that the dual is cyclic of order n. Let $\chi\colon G \to \mathbb{C}$ be a character. Since the group is cyclic, the character is entirely determined by the value it takes on the generator g. Since $g^n = e$, obviously $\chi(g)^n = 1$, so $\chi(g)^n \in \mu_n = \{\exp(2k\pi i/n) : k = 0, 1, \dots, n-1\}$. For each possible choice of $\chi(g) \in \mu_n$ we get a different character. It is then easy to see that $\chi\colon G \to \mathbb{C}$ such that $\chi(g) = \exp(2\pi i/n)$ generates this group, i.e. $\hat{G} = \left\{\chi, \chi^2, \dots, \chi^{n-1}, \chi^n = e_{\hat{G}}\right\}$.

Now since every finite abelian group can be written as a finite product of finite cycles, all we need to do is check whether the statement holds for a product of two groups A, B that are themselves self-dual, i.e. $\widehat{A \times B} \cong \widehat{A} \times \widehat{B}$. This is self-evident as every character $\chi \in \widehat{A \times B}$ is determined by its restrictions to $A \times \{e_B\}$ and $\{e_A\} \times B$, since

$$\chi(a, b) = \chi((a, e_B)(e_A, b)) = \chi(a, e_B)\chi(e_A, b).$$

Thus $\chi \leftrightarrow (\chi|_A, \chi|_B)$ is an isomorphism. □

2.2 C^*-algebras

In the following section we will introduce some notions stemming from the theory of C^*-algebras. We assume the Reader is familiar with the notion of a C^*-algebra. For reference, see Blackadar [1].

Multiplication in a C^*-algebra A is a bilinear map

$$m\colon A \times A \to A.$$

By the universal property of the tensor product, m extends uniquely to a linear map

$$m^*\colon A \otimes A \to A.$$

Throughout this thesis, we will use the same notation (m) for both the bilinear multiplication and its unique linear extension, relying on the context to distinguish between the two.

Definition 2.2.1 (Linear functional)**.** A *linear functional* on a C^*-algebra A is a linear function $\psi\colon A \to \mathbb{C}$.

Remark 2.2.1. We will call linear functionals simply as *functionals*.

The following notions will be useful for defining an inner product on C^*-algebras. Positiveness and faithfulness are all we need to define a norm on a finite-dimensional C^*-algebra equipped with a suitable functional.

Definition 2.2.2 (Positive functional)**.** A functional ψ on a C^*-algebra A is called *positive*, if

$$\psi(a^*a) \geq 0$$

for any $a \in A$.

Definition 2.2.3 (Faithful functional)**.** A positive functional ψ on a C^*-algebra A is called *faithful*, if

$$\psi(a^*a) = 0 \Leftrightarrow a = 0$$

for any $a \in A$.

Remark 2.2.2. In the literature, the functional ψ is often called a *state*.

The following notion is akin to the notion of a group action on a set.

Definition 2.2.4 (Group action on a C^*-algebra)**.** By *group action* of a group G on a C^*-algebra A we mean a group homomorphism

$$\alpha\colon G \to \mathrm{Aut}(A).$$

We will write $g \cdot a$ instead of $\alpha(g)(a)$ then the action in question is clear.

2.2.1 Crossed-product C^*-algebra

In this section we will introduce the notion of a crossed-product C^*-algebra. We will do so in a limited manner, only to the extent to which the theory of such products is necessary for our purposes. For a more general reference we refer the Reader to Blackadar [1].

The following definitions are based off of Sierakowski [6].

Definition 2.2.5 (C^*-dynamical system)**.** A C^*-dynamical system is a triple (A, G, α), where A is a C^*-algebra, G is a group, and $\alpha\colon G \to \mathrm{Aut}(A)$ is a group action on A. For a C^* dynamical system (A, G, α) with G finite we define

$$C_c(G, A, \alpha) = \{f\colon G \to A\colon\ \mathrm{supp}(f) \text{ is finite}\},$$

i.e. the space of all finitely supported functions on G with values in A.

$C_c(G, A, \alpha)$ is equipped with both a product and a $*$-operation in such a way that the action becomes "inner", i.e. $g \cdot a = u_g a u_g^*$ for all $g \in G$, $a \in A$. More precisely we define those two operations in the following way:

$$ab = \sum_{g,h \in G}$$

Definition 2.2.6 (Covariant representation)**.** Fix a dynamical system (A, G, α). A *covariant representation* (π, u, H) of (A, G, α) consist of a unitary representation $u\colon G \to B(\mathcal{H})$ of G and a representation $\pi\colon A \to B(\mathcal{H})$ of A such that $u(g)\pi(a)u(g)^* = \pi(g \cdot a)$ for every $g \in G$ and $a \in A$.

Definition 2.2.7 (Full C^*-algebra norm on $C_c(G, A, \alpha)$)**.** The full C^*-algebra norm on $C_c(G, A, \alpha)$ is given by

$$\|\cdot\| = \sup \|(\pi \times u)(\cdot)\|,$$

where the supremum is taken over all covariant representations (π, u, H) of (A, G, α).

Definition 2.2.8 (Crossed product C^*-algebra)**.** The crossed product of a C^*-algebra and a discrete group G is the completion of $C_c(G, A, \alpha)$ with respect to the full norm. We denote the crossed product by $\tilde{B} \rtimes_{\hat{\alpha}} \hat{G}$.

2.3 Matrix Lie Groups and Algebras

In this section we will remind the Reader some notions pertaining to the theory of matrix Lie groups and algebras that will be relevant to the rest of the thesis. A *Lie group* is a group that is also a smooth differential manifold such that the group action and taking inverses are smooth maps. A *Lie algebra* $\mathfrak{g}$ is a vector space over any field (in our case over $\mathbb{C}$), equipped with a bilinear form (often called the Lie bracket)

$$[\cdot,\cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$$

that is antisymmetric and has the Jacobi property:

- $[x, y] = -[y, x]$,
- $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$.

Those two notions are closely related, as Lie algebras naturally arise as tangent spaces of the identity element of Lie groups.

We will be concerned with *matrix Lie groups*, namely subgroups of the general linear group

$$GL(n, \mathbb{C}) = \{A \in M_n(\mathbb{C}) : \det(A) \neq 0\},$$

equipped with matrix multiplication. Examples include the unitary group

$$U(n) = \{U \in M_n(\mathbb{C}) : U^*U = UU^* = I\},$$

as well as its subgroup, the special unitary group

$$SU(n) = \{U \in M_n(\mathbb{C}) : U^*U = UU^* = I,\ \det(U) = 1\},$$

and the projective unitary group

$$PU(n) = U(n)/Z(U(n)),$$

where $Z(U(n))$ denotes the center of $U(n)$. In all these cases the respective Lie group possess the subspace topology inherited from $M_n(\mathbb{C})$, or, in the case of $PU(n)$, the quotient topology.

Proposition 2.3.1. $Z(U(n)) \cong \{\lambda \in \mathbb{C} : |\lambda| = 1\}$.

Proof. Recall that the center of $U(n)$ is defined as

$$Z(U(n)) = \{A \in U(n) : \forall U \in U(n)\ AU = UA\}.$$

We will show that

$$Z(U(n)) = \{\lambda I_n : |\lambda| = 1\}.$$

Let $A \in Z(U(n))$. Then A commutes with every unitary matrix. In particular, it commutes with every permutation matrix. Let P_{ij} denote the permutation matrix that exchanges the i-th and j-th basis vectors. Since

$$AP_{ij} = P_{ij}A,$$

conjugation by P_{ij} leaves A unchanged. As conjugation by P_{ij} exchanges the i-th and j-th rows and columns of A, it follows that all off-diagonal entries of A must vanish and all diagonal entries must be equal. Hence,

$$A = \lambda I_n$$

for some $\lambda \in \mathbb{C}$.

Since $A \in U(n)$, we have

$$A^* A = I.$$

Substituting $A = \lambda I_n$, we obtain

$$(\overline{\lambda} I_n)(\lambda I_n) = |\lambda|^2 I_n = I_n,$$

which implies

$$|\lambda| = 1.$$

Conversely, let $A = \lambda I_n$ with $|\lambda| = 1$. Then $A \in U(n)$, and for every $U \in U(n)$,

$$AU = \lambda U = UA.$$

Therefore $A \in Z(U(n))$.

We conclude that

$$Z(U(n)) = \{\lambda I_n : |\lambda| = 1\},$$

as claimed. □

We will also be interested in

$$SO(n) = \left\{ R \in M_n(\mathbb{R}) : R^T R = RR^T = I, \det(R) = 1 \right\},$$

i.e. the Lie group of orientation-preserving rotations around the origin point.

For matrix Lie algebras, the Lie bracket is given by the matrix commutator

$$[X, Y] = XY - YX,$$

for matrices X and Y in the Lie algebra.

The Lie algebra corresponding to $GL(n, \mathbb{C})$ is denoted by $\mathfrak{gl}(n, \mathbb{C})$ and consists of all complex $n \times n$ matrices,

$$\mathfrak{gl}(n, \mathbb{C}) = M_n(\mathbb{C}).$$

On the other hand the Lie algebra corresponding to $SU(n)$ is denoted by $\mathfrak{su}(n)$ and consists of all traceless skew-Hermitian matrices,

$$\mathfrak{su}(n) = \{X \in M_n(\mathbb{C}) : X^* = -X,\ \mathrm{Tr}(X) = 0\}.$$

We can treat $\mathfrak{su}_2$ as a 3-dimensional vector space (why?).

Chapter 3

Quantum sets and quantum graphs

In this chapter we will introduce the notions of quantum sets and quantum graphs. They were first considered by...

3.1 Quantum sets

Let B be a C^*-algebra equipped with a faithful positive functional ψ. By the Gelfand-Naimark theorem, B can be realized concretely as a closed $*$-subalgebra of the bounded linear operators $B(\mathcal{H})$ on some Hilbert space $\mathcal{H}$. When B is finite-dimensional, this abstract framework simplifies considerably: we can identify B with its own Hilbert space completion $L^2(B, \psi)$ by defining the inner product directly via the functional:

$$\langle x, y \rangle = \psi(x^* y) \quad \forall x, y \in B.$$

The following definition introduces one of the central notions used in this thesis. It is taken directly from Schäfer i Tobolski [5].

Definition 3.1.1. (Quantum set, Schäfer i Tobolski [5, Definition 2.1]) A quantum set (B, ψ) is a pair consisting of a finite-dimensional C^*-algebra B and a faithful positive functional $\psi\colon B \to \mathbb{C}$ such that

$$mm^* = \mathrm{Id}_B$$

holds for the multiplication map $m\colon B \otimes B \to B$ and its adjoint m^* given by

$$\langle m(a \otimes b), c \rangle = \langle a \otimes b, m^*(c) \rangle_{\psi \otimes \psi}$$

for all $a, b, c \in B$.

Remark 3.1.1. We will call any functional with the property given in the definition above a *quantum set functional* on B.

Remark 3.1.2. Recall that in general, a *comultiplication* on a C^*-algebra is a $*$-homomorphism

$$\Delta \to A \otimes A$$

such that the following diagram commutes:

$$\begin{tikzcd} A \arrow[r, "\Delta"] \arrow[d, "\Delta"'] & A \otimes A \arrow[d, "\Delta\otimes\mathrm{id}"] \\ A \otimes A \arrow[r, "\mathrm{id}\otimes\Delta"'] & A \otimes A \otimes A \end{tikzcd}$$

Note that in general a comultiplication is *not* unique. In our setting it is uniquely determined by the additional structure and the Riesz representation theorem for Hilbert spaces.

Remark 3.1.3. Since in our setup the comultiplication is given by the scalar product and regular multiplication m, we will denote it by m^*.

Example 3.1.1. Every finite set X gives rise to a quantum set $(C(X), \psi_X)$ with $\psi_X \colon C(X) \to \mathbb{C}$, $\psi_X(f) = \sum_{x\in X} f(x)$ for any $f \in C(X)$. Let X be a finite set and $C(X)$ be the $*$-algebra of complex-valued functions on X, where the involution is given by pointwise complex conjugation: $f^*(x) = \overline{f(x)}$.

To show that ψ_X is positive, we must verify that $\psi_X(f^*f) \geq 0$ for all $f \in C(X)$. For any function f, the product f^*f evaluates pointwise to:

$$(f^*f)(x) = \overline{f(x)}f(x) = |f(x)|^2.$$

Applying the functional yields:

$$\psi_X(f^*f) = \sum_{x\in X} |f(x)|^2.$$

Thus the functional is positive.

To prove faithfulness, suppose $\psi_X(f^*f) = 0$ for some $f \in C(X)$. This implies:

$$\sum_{x\in X} |f(x)|^2 = 0.$$

Because each term $|f(x)|^2$ is strictly non-negative, a sum of such terms can only equal zero if every individual term is zero. Thus, we must have:

$$|f(x)|^2 = 0 \quad \forall x \in X,$$

which directly implies $f(x) = 0$ for all $x \in X$. Therefore, f is the zero function ($f = 0$), establishing that ψ_X is faithful.

Let $\{e_x\}_{x\in X}$ be the standard orthonormal basis of characteristic functions on X. Recall the definitions of the multiplication map m and its adjoint m^*:

$$m(f \otimes g) = fg,$$
$$m^*(e_x) = e_x \otimes e_x.$$

We wish to show that the composition $mm^*\colon C(X) \to C(X)$ is the identity operator (Id). Let us evaluate the action of mm^* on an arbitrary basis element e_x:

$$(mm^*)(e_x) = m\big(m^*(e_x)\big) = m(e_x \otimes e_x) = e_x e_x = e_x.$$

Because the linear operators mm^* and Id agree on every element of the basis $\{e_x\}_{x\in X}$, by linearity they must be equal on the entire space $C(X)$. Therefore:

$$mm^* = \mathrm{Id}.$$

Example 3.1.2. In particular, $(\mathbb{C}^n, \psi_n)$ with $\psi_n(e_i) = 1$ is a quantum set. The functional naturally extends to:

$$\psi_n(x) = \sum_{i=1}^{n} x_i.$$

To show that ψ_n is positive, we evaluate it on the element x^*x. Using the orthogonality of the basis, we have:

$$x^*x = \left(\sum_{i=1}^{n} \overline{x_i} e_i\right)\left(\sum_{j=1}^{n} x_j e_j\right) = \sum_{i,j=1}^{n} \overline{x_i} x_j e_i e_j = \sum_{i=1}^{n} \overline{x_i} x_i e_i = \sum_{i=1}^{n} |x_i|^2 e_i.$$

Applying the functional yields:

$$\psi_n(x^*x) = \psi_n\left(\sum_{i=1}^{n} |x_i|^2 e_i\right) = \sum_{i=1}^{n} |x_i|^2 \psi_n(e_i) = \sum_{i=1}^{n} |x_i|^2.$$

Since $|x_i|^2 \geq 0$ for all i, the sum of these non-negative real numbers is also non-negative, meaning $\psi_n(x^*x) \geq 0$. Thus, the functional is positive.

To prove faithfulness, suppose $\psi_n(x^*x) = 0$ for some $x \in \mathbb{C}^n$. This implies:

$$\sum_{i=1}^{n} |x_i|^2 = 0.$$

Because each term $|x_i|^2$ is strictly non-negative, the sum can only be zero if every individual component satisfies $|x_i|^2 = 0$ for all $i \in \{1, \ldots, n\}$. This directly implies $x_i = 0$ for all i. Therefore, $x = \sum_{i=1}^{n} 0 \cdot e_i = 0$, establishing that ψ_n is faithful.

The adjoint map $m^*\colon \mathbb{C}^n \to \mathbb{C}^n \otimes \mathbb{C}^n$ acts on a basis element e_i by:

$$m^*(e_i) = e_i \otimes e_i.$$

We wish to show that the composition $mm^*\colon \mathbb{C}^n \to \mathbb{C}^n$ is the identity operator. We check:

$$(mm^*)(e_i) = m\big(m^*(e_i)\big) = m(e_i \otimes e_i) = e_i.$$

Because the linear operators mm^* and Id agree on every element of the basis $\{e_i\}_{i=1}^{n}$, by linearity they must be equal on the entire vector space $\mathbb{C}^n$. Therefore:

$$mm^* = \mathrm{Id}.$$

Example 3.1.3. For any $n \in \mathbb{N}_+$, $(M_n, n\,\mathrm{Tr})$ is a quantum set, with Tr being the trace. We will later show that this structure can be obtained as the crossed product $(\mathbb{C}^n, \psi_n)$ and $\hat{\mathbb{Z}}_n$ with the standard action, i.e.

$$(M_n, n\,\mathrm{Tr}) \cong (\mathbb{C}^n \rtimes \hat{\mathbb{Z}}_n, \psi_\rtimes).$$

This is proven in 3.1.2.

Lemma 3.1.1 (Comultiplication for matrix algebras)**.** *In the context of* $(M_n, n\,\mathrm{Tr})$*, the comultiplication is given by the formula*

$$m^*(e_{ij}) = \sum_{k=1}^{n} e_{ik} \otimes e_{kj}$$

with e_{ij} *being the matrix with a single* 1 *in the* i*-th row and* j*-th column, and extended linearly.*

Proof. This is simply a computational excercise. We will use the following well-known formulas:

1. $e_{ab}e_{cd} = \delta_{bc}e_{ad}$,
2. $\langle e_{ij}, e_{kl}\rangle = \mathrm{Tr}(e_{ji}e_{kl}) = \delta_{ik}e_{jl}$,

with

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases}$$

In particular $\{e_{ij}\}_{i,j=1,\dots,n}$ form a orthonormal basis.

Take arbitrary basis vectors e_{ab}, e_{cd}. Then

$$m(e_{ab} \otimes e_{cd}) = e_{ab}e_{cd} = \delta_{bc}e_{ad},$$

hence

$$\begin{aligned} \langle m(e_{ab} \otimes e_{cd}), e_{ij}\rangle &= \delta_{bc}\langle e_{ad}, e_{ij}\rangle \\ &= \delta_{bc}\delta_{ai}\delta_{dj}. \end{aligned}$$

Now compute the inner product with the proposed adjoint:

$$\begin{aligned} \left\langle e_{ab} \otimes e_{cd}, \sum_{k=1}^{n} e_{ik} \otimes e_{kj} \right\rangle &= \sum_{k=1}^{n} \langle e_{ab}, e_{ik}\rangle \langle e_{cd}, e_{kj}\rangle \\ &= \sum_{k=1}^{n} \delta_{ai}\delta_{bk}\delta_{ck}\delta_{dj} \\ &= \delta_{ai}\delta_{bc}\delta_{dj}. \end{aligned}$$

The two expressions agree. Since the matrix unix form an orthonormal basis, this linearly extends to all matrices. □

3.1.1 Crossed product quantum sets

Schäfer i Tobolski [5] have introduced the notion of crossed product quantum sets. Suppose we are given a finite-dimensional C^*-algebra $\tilde{B}$ along with a dual group action $\hat{a}\colon \hat{G} \to \operatorname{Aut}(\tilde{B})$ where G is finite and abelian. One can then consider the crossed product

$$B := \tilde{B} \rtimes_{\hat{\alpha}} \hat{G}.$$

If $\tilde{B}$ is equipped with a faithful functional $\tilde{\psi}$ in such a way as to make $(\tilde{B}, \tilde{\psi})$ a quantum set, one can define a new faithful functional on B to make it a quantum set, as described below.

Definition 3.1.2. (Crossed product quantum set, Schäfer i Tobolski [5, Definition 3.1]) Let $(\tilde{B}, \tilde{\psi})$ be a quantum set and let $\hat{\alpha}\colon \hat{G} \to \operatorname{Aut}(\tilde{B}, \tilde{\psi})$ be an action of the dual of a finite abelian group G. The *crossed product quantum set* is defined as

$$(\tilde{B}, \tilde{\psi}) \rtimes_{\hat{\alpha}} \hat{G} := \left(\tilde{B} \rtimes_{\hat{\alpha}} \hat{G}, \psi_{\rtimes}\right),$$

where $\tilde{B} \rtimes_{\hat{\alpha}} \hat{G}$ is the crossed product C^*-algebra and $\psi_{\rtimes}$ is the functional given by

$$\psi_{\rtimes}\left(\sum_{\chi \in \hat{G}} b_\chi u_\chi\right) = n\tilde{\psi}(b_1)$$

with $n = |\hat{G}|$.

The proof that this definition is sensible, i.e. the newly defined object is indeed a quantum set, can be seen in Schäfer i Tobolski [5], Proposition 3.4.

Fact 3.1.1. *(***tobolksi26***, Example 3.5) If $\hat{\alpha} = \mathrm{triv}$, i.e. it is the trivial action, then*

$$\tilde{B} \rtimes_{\hat{\alpha}} \hat{\hat{G}} \cong \tilde{B} \otimes C(G),$$

where the isomorphism is given by $b_\chi u_\chi \mapsto b_\chi \otimes \tilde{u}_\chi$, where $\tilde{u}_\chi \in C(G)$ is the function that evaluates to $\chi(g)$ on $g \in G$.

Lemma 3.1.2. *Let $\hat{\alpha}\colon \hat{\mathbb{Z}}_n \to A$ be the action of cyclically shifting elements. Then*

$$(M_n, n\operatorname{Tr}) \cong (\mathbb{C}^n \rtimes_\alpha \hat{\mathbb{Z}}_n, \psi_{\rtimes}).$$

3.2 Quantum graphs

A simple graph is classically defined as a pair (V, E) with V serving as the *set of vertices* and $E \subseteq [V]^2$ serving as the *set of edges*. Such a graph could be interpreted as a matrix $A = [a_{ij}] \in \{0,1\}^{n\times n}$ with $\{v_i, v_j\} \in E \Leftrightarrow a_{ij} = 1$; such a representation is often called the *adjacency matrix*. The key property of such matrices is that they are invariant with respect to entrywise multiplication. In turn we can use this property to generalise graphs on sets to graphs on quantum sets.

The following definitions are due to Schäfer i Tobolski [5].

Definition 3.2.1. (Quantum adjacency matrix, Schäfer i Tobolski [5, Definition 2.5]) Let (B, ψ) be a quantum set. A *quantum adjacency matrix* on this quantum set is a linear map $A\colon B \to B$ which is Schur-idempotent and *-preserving, by which we mean that it satisfies

$$m(A \otimes A)m^* = A, \qquad A(b^*) = A(b)^* \quad \text{for all } b \in B.$$

Definition 3.2.2. (Quantum graph, Schäfer i Tobolski [5, Definition 2.5]) A *quantum graph* on (B, ψ) is given by a quantum adjacency matrix on this quantum set.

For all intents and purposes we will identify both notions and simply call an appropriate $A\colon B \to B$ both a quantum adjancenty matrix and a quantum graph.

Remark 3.2.1. There are other equivalent definitions of a quantum graph, namely those based on

- an *edge projection*, i.e. a projection $\tilde{A} \in C(X) \otimes C(X)^{\mathrm{op}}$, see Musto, Reutter i Verdon [4],
-

Definition 3.2.3. (Schäfer i Tobolski [5, Definition 4.6]) Let A be a quantum graph on a quantum set (B, ψ). We call the quantum graph

- *loopfree* if $m(A \otimes \mathrm{Id})m^* = 0$,
- *undirected* if $A = A^*$, where the adjoint is taken with respect to the inner product induced by ψ.

We say that a voltage quantum graph $\tilde{A} = \{\tilde{A}_g\}_{g \in G}$ has these properties if the same equations hold for $\sum_{g \in G} \tilde{A}_g$, respectively.

3.3 Voltage and Derived Quantum Graphs

3.3.1 Classical Voltage Graphs

Definition 3.3.1 (Classical voltage graph)**.** A *voltage graph* is a pair (Γ, λ) consisting of (not necesarily simple) finite directed graph Γ with vertex set V and edge set E, together with a labelling $\lambda : E \to G$ of the edges of Γ by elements of a finite group G.

Definition 3.3.2 (Classical derived graph)**.** The *derived graph* Γ^λ of (Γ, λ) is a graph with vertex set $V \times G$ and edge set $E \times G$. Whenever $e \in E$ is an edge from u to v in Γ and $g \in G$ is a group element, then the edge (e, g) goes from (u, g) to $(v, g\lambda(e))$ in Γ^λ.

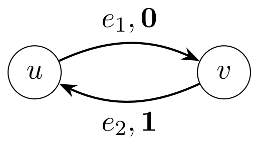

Figure 3.1: Base voltage graph Γ over $\mathbb{Z}_2$

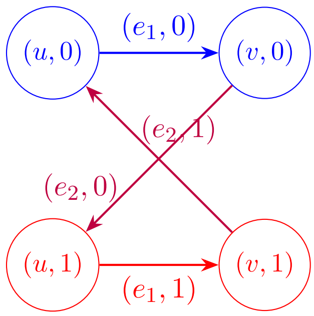

Figure 3.2: Derived graph Γ^λ (Directed 4-cycle)

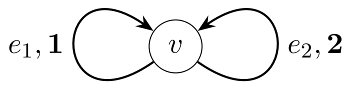

Figure 3.3: Base voltage graph Γ over $\mathbb{Z}_3$

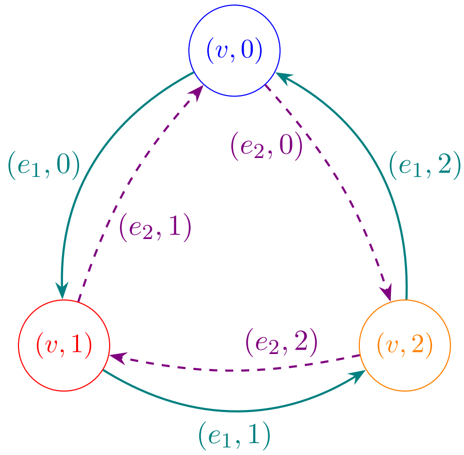

Figure 3.4: Derived graph Γ^λ (Two 3-cycles)

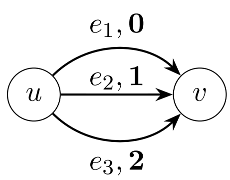

Figure 3.5: Base Theta graph Γ with edge labels

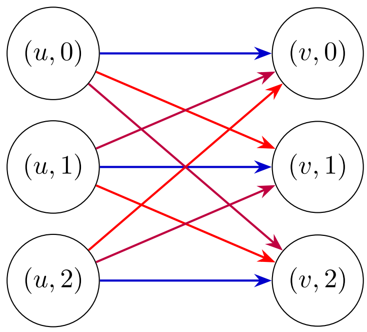

Figure 3.6: Derived graph Γ^λ ($K_{3,3}$ colored by edge class)

3.3.2 Generalisation to quantum voltage and derived quantum graphs

In the following definition we will introduce some key notions. The motivation behind the definition of X_g will be given later in the thesis.

Definition 3.3.3. (Schäfer i Tobolski [5], Definition 4.1) Let $(\tilde{B}, \tilde{\psi})$, $\hat{\alpha} : \hat{G} \to \mathrm{Aut}(\tilde{B}, \tilde{\psi})$, $(B, \psi_{\rtimes})$ and $(u_\chi)_{\chi \in \hat{G}} \subset B$ be as above.

1. For every $g \in G$ let $X_g := \frac{1}{n} \sum_{\chi \in \hat{G}} \overline{\chi(g)} u_\chi u_\chi^\dagger$, i.e.
$$X_g : B \to B, \quad b \mapsto \frac{1}{n} \sum_{\chi \in \hat{G}} \overline{\chi(g)} \langle u_\chi, b \rangle_{\psi_\rtimes} u_\chi.$$
2. A *voltage quantum graph* on the quantum set $(\tilde{B}, \tilde{\psi})$ with respect to the group action $\hat{\alpha} : \hat{G} \to \mathrm{Aut}(\tilde{B}, \tilde{\psi})$ is a family $\tilde{A} = (\tilde{A}_g)_{g \in G}$ of quantum adjacency matrices that is indexed by G and satisfies
$$\tilde{A}_g \hat{\alpha}_\chi = \hat{\alpha}_\chi \tilde{A}_g$$
for all $g \in G$ and $\chi \in \hat{G}$.
3. The *derived quantum graph* $\tilde{A} \rtimes_{\hat{\alpha}} \hat{G} : B \to B$ on $(B, \psi_\rtimes)$ is then given by
$$\tilde{A} \rtimes_{\hat{\alpha}} \hat{G} := \sum_{g \in G} m(X_g \otimes E_\rtimes^* \tilde{A}_g E_\rtimes) m^*,$$
where we write (here and in the rest of the section) m for the multiplication map on B, i.e. $m = m_\rtimes$.

One can easily check that X_g is a adjacency matrix for any $g \in G$.

Lemma 3.3.1. *If $A, B \in M_n(\{0,1\})$ and $m^* : \mathbb{C}^n \to \mathbb{C}^n \otimes \mathbb{C}^n$ is the adjoint of the multiplication map m, then $m(A \otimes B)m^* = A \odot B$, the entrywise (Hadamard) product of A and B.*

Proof. Let $\{e_1, \ldots, e_n\}$ be the standard basis of $\mathbb{C}^n$. The componentwise multiplication map m acts as:
$$m(e_i \otimes e_j) = \delta_{ij} e_i$$
By definition of the adjoint via the standard inner product, $\langle m(e_i \otimes e_j), e_k \rangle = \langle e_i \otimes e_j, m^*(e_k) \rangle$. Since the left-hand side is $\delta_{ij}\delta_{ik}$, it follows that:
$$m^*(e_k) = e_k \otimes e_k$$
We expand the action of the matrices A and B on a basis vector e_k via their entries A_{ik}, B_{jk}:
$$Ae_k = \sum_{i=1}^{n} A_{ik} e_i, \quad Be_k = \sum_{j=1}^{n} B_{jk} e_j$$

Applying the composite operator $m(A \otimes B)m^*$ to e_k yields:

$$\begin{aligned} m(A \otimes B)m^*(e_k) &= m(Ae_k \otimes Be_k) \\ &= m\left(\sum_{i=1}^{n}\sum_{j=1}^{n} A_{ik}B_{jk}(e_i \otimes e_j)\right) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} A_{ik}B_{jk}\delta_{ij}e_i \\ &= \sum_{i=1}^{n}(A_{ik}B_{ik})e_i \end{aligned}$$

Thus, the (i,k)-th entry of the matrix $m(A \otimes B)m^*$ is exactly $A_{ik}B_{ik}$. Because A and B are binary, this entry evaluates to 1 if and only if $A_{ik} = 1$ and $B_{ik} = 1$, and 0 otherwise. □

Remark 3.3.1. $m(A \otimes B)m^*$ can be thought of as taking the intersecion of two quantum matrices blabla...

3.4 Isomorphisms and Quantum Isomorphisms of Quantum Sets

Definition 3.4.1. (Quantum isomorphisms, Schäfer i Tobolski [5, Definition 2.8]) Let (B_1, ψ_1) and (B_2, ψ_2) be quantum sets and let $A_1 \colon B_1 \to B_1$ and $A_2 \colon B_2 \to B_2$ be quantum graphs on (B_1, ψ_1), (B_2, ψ_2), respectively.

- The two quantum sets are *isomorphic* if there is a $*$-isomorphism $\phi \colon B_1 \to B_2$ such that $\psi_2\phi = \psi_1$. We denote by $\mathrm{Aut}(B, \psi)$ the group of quantum set automorphisms of a quantum set (B, ψ).

- The two quantum graphs are *isomorphic* if there is an isomorpshism $\phi \colon B_1 \to B_2$ of the underlying quantum sets such that $\phi A_1 = A_2\phi$. We denote by $\mathrm{Aut}(A)$ the group of quantum graph automorphisms of a quantum graph A.

- The two quantum sets are *quantum isomorphic*, written $(B_1, \psi_1) \cong (B_2, \psi_2)$, if there is a finite-dimensional Hilbert space $\mathcal{H}$ and a unital $*$-homomorphism

  $$\rho \colon B_1 \to B_2 \otimes B(\mathcal{H}),$$

  such that the map

  $$p \colon L^2(B_1) \otimes \mathcal{H} \to L^2(B_2) \otimes \mathcal{H},\ a \otimes \xi \mapsto \rho(a)(1 \otimes \xi)$$

  is unitary as an operator between Hilbert spaces.

- The two quantum graphs are *quantum isomorphic*, written $A_1 \cong_q A_2$, if there is a quantum isomorphism $\rho\colon B_1 \to B_2 \otimes B(\mathcal{H})$ between the underlying quantum sets such that
$$\rho A_1 = (A_2 \otimes \mathrm{Id}_{B(\mathcal{H})})\rho$$
holds.

Chapter 4

Quantum graphs over 3×3 matrices

The classification on 2×2 matrices was done by Matsuda [3].

4.1 Pauli Matrices and their generalisations

4.1.1 Pauli matrices

An important basis of the Lie algebra $\mathfrak{sl}(2, \mathbb{C})$ is given by the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The Pauli matrices are Hermitian, traceless, and linearly independent. Since $\dim_{\mathbb{C}} \mathfrak{sl}(2, \mathbb{C}) = 3$, they form a basis of $\mathfrak{sl}(2, \mathbb{C})$.

Since the Pauli matrices are Hermitian rather than skew-Hermitian, they do not belong to the Lie algebra $\mathfrak{su}(2)$. Instead, the matrices

$$i\sigma_1, \qquad i\sigma_2, \qquad i\sigma_3$$

form a basis of $\mathfrak{su}(2)$, as they are traceless and skew-Hermitian.

4.1.2 Generalised Pauli matrices

For dimensions greater than two, the role of the Pauli matrices is played by the *generalised Pauli matrices*, also known as the *clock* and *shift* matrices. In dimension three they are defined by

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

where

$$\omega = \exp(2\pi i/3).$$

The matrices P and Q are unitary and satisfy

$$P^3 = Q^3 = I,$$

together with the commutation relation

$$PQ = \omega QP.$$

The clock and shift matrices are related by the discrete Fourier transform. The Fourier matrix is defined by

$$F = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}.$$

It is unitary and satisfies

$$F^*F = FF^* = I.$$

The Fourier transform exchanges the roles of the clock and shift operators according to

$$FPF^* = Q^{-1}, \qquad FQF^* = P.$$

Therefore, the matrices P and Q are unitarily equivalent and represent two complementary bases in the three-dimensional Hilbert space. This duality is the finite-dimensional analogue of the relation between position and momentum operators and is a fundamental property of the generalised Pauli matrices.

Lemma 4.1.1. *The matrices*

$$P^aQ^b, \qquad a, b \in \{0, 1, 2\},$$

form an orthogonal basis of $M_3(\mathbb{C})$ with respect to the Hilbert–Schmidt inner product

$$\langle A, B\rangle = \mathrm{Tr}(A^*B).$$

Proof. There are $3^2 = 9$ matrices of the form P^aQ^b, while $\dim_{\mathbb{C}} M_3(\mathbb{C}) = 9$. It therefore suffices to show that they are pairwise orthogonal.

Since

$$P^* = P^{-1}, \qquad Q^* = Q^{-1},$$

we have

$$(P^aQ^b)^* = Q^{-b}P^{-a}.$$

Hence

$$\langle P^aQ^b, P^cQ^d\rangle = \mathrm{Tr}\left(Q^{-b}P^{-a}P^cQ^d\right) = \mathrm{Tr}\left(Q^{-b}P^{c-a}Q^d\right).$$

Using the commutation relation

$$Q^k P^\ell = \omega^{-k\ell} P^\ell Q^k,$$

we obtain

$$Q^{-b} P^{c-a} = \omega^{b(c-a)} P^{c-a} Q^{-b},$$

and therefore

$$\langle P^a Q^b, P^c Q^d \rangle = \omega^{b(c-a)} \operatorname{Tr}\left(P^{c-a} Q^{d-b}\right).$$

If $b \neq d$, then Q^{d-b} is a nontrivial permutation matrix, so $P^{c-a}Q^{d-b}$ has zero diagonal entries. Hence

$$\operatorname{Tr}(P^{c-a} Q^{d-b}) = 0.$$

If $b = d$, then

$$\langle P^a Q^b, P^c Q^b \rangle = \operatorname{Tr}(P^{c-a}).$$

Since

$$P^{c-a} = \operatorname{diag}(1, \omega^{c-a}, \omega^{2(c-a)}),$$

we have

$$\operatorname{Tr}(P^{c-a}) = 1 + \omega^{c-a} + \omega^{2(c-a)} = \begin{cases} 3, & a = c, \\ 0, & a \neq c, \end{cases}$$

because $1 + \omega + \omega^2 = 0$.

Therefore

$$\langle P^a Q^b, P^c Q^d \rangle = 3\delta_{ac}\delta_{bd},$$

which proves that the matrices $P^a Q^b$ are pairwise orthogonal. Since there are nine of them, they form an orthogonal basis of $M_3(\mathbb{C})$. □

4.2 Gromada's theorem

In the following section we will state...

The following theorem is due to Gromada [2].

Theorem 4.2.1 (Gromada [2], Theorem 3.7)**.** *Consider $m \in \{0, 1, \ldots, n^2\}$. Quantum graphs on M_n with m quantum edges are in a one-to-one correspondence with m-dimensional subspaces $V \subseteq \mathfrak{gl}_n$. The adjacency matrix is given by $A_V = \sum_{s=1}^m \xi_s \otimes \xi_s^*$, where $(\xi_1, \xi_2, \ldots, \xi_m)$ is some orthonormal basis of V.*

The quantum graph has no loops if and only if $V \subseteq \mathfrak{sl}_n$. It is undirected if and only if $V = V^*$.

Let $V, W \subseteq \mathfrak{gl}_n$ be two vector spaces corresponding to isomorphic quantum graphs. Then $V = UWU^*$ for some $U \in SU_3$.

Proof. See Gromada's paper for reference. □

Lemma 4.2.1 (Gromada [2], Lemma 3.8)**.** *Any vector subspace $V \subseteq \mathfrak{gl}_n$ with $V = V^*$ has a self-adjoint basis $\{\xi_s\}$, $\xi_s = \xi_s^*$.*

Proof. We construct the desired basis inductively. Suppose we have obtained a linearly independent set of self-adjoint vectors $\{\xi_s\}$ spanning a subspace $V_1 \subseteq V$. By construction, V_1 is invariant under the involution, meaning $V_1 = V_1^*$.

If $V_1 \neq V$, let $V_2 = V_1^\perp$ be the orthogonal complement of V_1 with respect to the inner product. Because the inner product and the involution are compatible, V_2 is also invariant under the map $\cdot^*$.

Select any non-zero element $\xi \in V_2$ and decompose it into its real and imaginary self-adjoint components:

$$\xi_{\mathrm{Re}} = \frac{\xi + \xi^*}{2} \quad \text{and} \quad \xi_{\mathrm{Im}} = \frac{i(\xi^* - \xi)}{2}$$

Both ξ_{Re} and ξ_{Im} are self-adjoint elements belonging to V_2. Since $\xi \neq 0$, at least one of these components must be non-zero. We then extend our independent set by appending either both components (if they are linearly independent) or the sole non-zero component (if they are dependent). Iterating this process eventually yields a full self-adjoint basis for V. □

Remark 4.2.1. The following lemma tells us that any undirected graph on M_n can be thought of as being made from symmetric quantum edges only.

Lemma 4.2.2. *A complex vector space $V = V^* \subseteq \mathfrak{gl}_n$ uniquely determines a real vector space spanned by the basis $\{\xi_s\}$ of self-adjoint elements $\xi_s = \xi_s^*$.*

Proof. Let $V \subseteq \mathfrak{gl}_n$ be a complex vector space satisfying

$$V = V^*.$$

Since V is invariant under taking the Hermitian adjoint, every element $v \in V$ admits the decomposition

$$v = \frac{v + v^*}{2} + i\frac{v - v^*}{2i}.$$

Both summands belong to V, because V is closed under addition, scalar multiplication, and the adjoint operation. Moreover,

$$\left(\frac{v + v^*}{2}\right)^* = \frac{v + v^*}{2},$$

and

$$\left(i\frac{v - v^*}{2i}\right)^* = i\frac{v - v^*}{2i},$$

so both matrices are self-adjoint.

Now let $\xi \in V$ be self-adjoint. Let $\{\xi_s\}$ be a self-adjoint basis. Then there exist coefficients $c_s \in \mathbb{C}$ such that

$$\xi = \sum_s c_s \xi_s.$$

Taking the Hermitian adjoint yields

$$\xi = \xi^* = \sum_s \overline{c_s} \xi_s,$$

because each ξ_s is self-adjoint. By the uniqueness of the expansion with respect to the basis $\{\xi_s\}$, we obtain

$$c_s = \overline{c_s}$$

for every s. Hence each coefficient c_s is real.

Therefore every self-adjoint element of V is a real linear combination of the basis elements $\{\xi_s\}$. Consequently, the self-adjoint part of V is a real vector space uniquely determined by V. $\square$

By the above lemma we can think of self-adjoint $V \subseteq \mathfrak{sl}_n$ as real subspaces of $\mathfrak{sl}_n$.

Remark 4.2.2. A self-adjoint subspace $V \subseteq \mathfrak{sl}_3$ generated by the generalized Pauli matrices can only be of even dimension, i.e., $\dim(V) = m \in \{0, 2, 4, 6, 8\}$. This is because the non-identity Pauli basis generators $P^a Q^b \in \mathcal{B}$ are non-self-adjoint (as $(a, b) \neq (3 - a, 3 - b)$ in $\mathbb{Z}_3 \times \mathbb{Z}_3$) and are partitioned into mutually disjoint pairs of adjoint partners $\{v, v^*\}$. Since $V = V^*$, the inclusion of any generator $v \in V$ forces its linearly independent reflection partner $v^* \in V$, meaning V decomposes into a direct sum of 2-dimensional reflection-invariant blocks $\mathcal{P}_k = \mathrm{span}\{v, v^*\}$. Thus, $\dim(V) = 2r$ for some integer $r \geq 0$.

To characterize all possible loop-free and undirected quntum graphs we only need to focus on the relevant subspaces.

4.3 Classification of Self-Reflective Subspaces Generated by P and Q

4.3.1 Fundamental Reflection Invariant Pairs

Since P and Q are unitary matrices ($P^* = P^2$ and $Q^* = Q^2$), the conjugate transpose of an arbitrary generator obeys:

$$(P^a Q^b)^* = \omega^{-ab} P^{3-a} Q^{3-b}.$$

Because any self-reflective subspace must be closed under conjugate transposition, every basis generator $P^a Q^b$ must belong to a subspace alongside its structural reflection partner $P^{3-a} Q^{3-b}$. This partitions the 8 non-identity generators into 4 mutually exclusive, reflection-invariant blocks of dimension 2:

$$\begin{aligned}
\mathcal{P}_1 &= \{P, P^2\} && \text{(Purely Diagonal Block)}\\
\mathcal{P}_2 &= \{Q, Q^2\} && \text{(Purely Shift Block)}\\
\mathcal{P}_3 &= \{PQ, P^2Q^2\} && \text{(First Mixed Block)}\\
\mathcal{P}_4 &= \{P^2Q, PQ^2\} && \text{(Second Mixed Block)}
\end{aligned}$$

4.3.2 Action of the Discrete Fourier Transform

Let F denote the 3×3 Discrete Fourier Transform matrix. The unitary similarity transformation $X \mapsto FXF^*$ acts as an automorphism on the generalized Pauli algebra, mapping clock and shift operators into one another:

$$FPF^* = Q^2 \quad \text{and} \quad FQF^* = P.$$

On the fundamental blocks $\mathcal{P}_k$, the DFT induces a natural pair-swapping involution:

$$F\mathcal{P}_1F^* = \mathcal{P}_2, \quad F\mathcal{P}_2F^* = \mathcal{P}_1, \quad F\mathcal{P}_3F^* = \mathcal{P}_4, \quad F\mathcal{P}_4F^* = \mathcal{P}_3.$$

4.3.3 Complete Classification Theorem

Because the fundamental building blocks $\mathcal{P}_k$ are non-overlapping 2-dimensional sets, every self-reflective subspace generated by powers of P and Q is formed by taking a direct sum of a subset of these blocks (including the empty set).

Theorem 4.3.1. *All self-reflective subspaces $V \subseteq M_3$ generated by P and Q are classified by their dimension $m = \dim(V)$ and their orbits under the Discrete Fourier Transform action $V \mapsto FVF^*$ as follows:*

1. ***Odd Dimensions** $m \in \{1, 3, 5, 7\}$**:** No such subspaces exist (0 subspaces).*
2. ***Dimension** $m = 0$ **(1 subspace):** The trivial zero space $V_0 = \{0\}$.*
   - *Orbit: $\{V_0\}$ (F-invariant).*
3. ***Dimension** $m = 2$ **(4 subspaces):** Let $V_k = \operatorname{span}(\mathcal{P}_k)$ for $k \in \{1, 2, 3, 4\}$. These partition into 2 Fourier orbits of size 2:*
   - ***Diagonal/Shift Orbit:** $\{V_1, V_2\}$, where $FV_1F^* = V_2$.*
   - ***Mixed-Shift Orbit:** $\{V_3, V_4\}$, where $FV_3F^* = V_4$.*
4. ***Dimension** $m = 4$ **(6 subspaces):** Let $W_{i,j} = \operatorname{span}(\mathcal{P}_i \oplus \mathcal{P}_j)$ for $1 \leq i < j \leq 4$. These partition into 4 Fourier orbits (2 invariant singletons and 2 pairs):*
   - ***Pure Block Orbit:** $\{W_{1,2}\}$ (F-invariant, since $F\mathcal{P}_1F^* = \mathcal{P}_2$).*
   - ***Mixed Block Orbit:** $\{W_{3,4}\}$ (F-invariant, since $F\mathcal{P}_3F^* = \mathcal{P}_4$).*

- ***Cross Orbit A:*** *$\{W_{1,3}, W_{2,4}\}$, where $FW_{1,3}F^* = W_{2,4}$.*
- ***Cross Orbit B:*** *$\{W_{1,4}, W_{2,3}\}$, where $FW_{1,4}F^* = W_{2,3}$.*

5. ***Dimension** $m = 6$ **(4 subspaces):** Let $U_k = \operatorname{span}\left(\bigoplus_{i\neq k} \mathcal{P}_i\right)$ for $k \in \{1,2,3,4\}$. These partition into 2 Fourier orbits of size 2:*
   - ***Complementary Diagonal/Shift Orbit:*** *$\{U_1, U_2\}$, where $FU_1F^* = U_2$.*
   - ***Complementary Mixed Orbit:*** *$\{U_3, U_4\}$, where $FU_3F^* = U_4$.*

6. ***Dimension** $m = 8$ **(1 subspace):** The full traceless algebra $\mathfrak{su}_3 = \operatorname{span}\left(\bigoplus_{k=1}^{4} \mathcal{P}_k\right)$.*
   - *Orbit: $\{\mathfrak{su}_3\}$ (F-invariant).*

Corollary 4.3.1. Including the trivial subspace V_0, there exist precisely 16 self-reflective subspaces generated by P and Q. Under the action of the Discrete Fourier Transform $V \mapsto FVF^*$, these 16 subspaces partition into *10 equivalence classes* (3 F-invariant fixed points and 7 swapped pairs).

4.4 Classification of different types of graphs

In this section we will consider the possibility of classification of quantum graphs that are not necessarily loopfree and/or are not undirected.

Theorem 4.4.1. *(Gromada [2, Theorem 3.11.]) A simple graph on M_2 is determined up to isomorphism only by the number of quantum edges $m \in \{0,1,2,3\}$.*

The proof of this theorem uses the fact that $PU_2 \cong SO_3$, which is well-known and is key in the proof. This cannot be generalised to higher dimensions, as the lemma below states.

Lemma 4.4.1. *The natural action*

$$PU(3) \curvearrowright \mathbb{P}(\mathbb{R}^8) = \{W \leq \mathbb{R}^8 : \dim(W) = 1\}$$

has infinitely many orbits.

Proof. Let

$$V = \mathfrak{su}(3) \cong \mathbb{R}^8,$$

and identify $\mathbb{P}(\mathbb{R}^8)$ with $\mathbb{P}(V)$. The action of $PU(3)$ on V is the adjoint representation

$$g \cdot X = gXg^{-1},$$

which induces an action on the 1-dimensional subspaces of V.

To prove that there are infinitely many orbits, it suffices to construct a projective invariant taking infinitely many values.

Define
$$q(X) = -\operatorname{tr}(X^2), \qquad c(X) = i\operatorname{tr}(X^3).$$
Since the trace is invariant under conjugation,
$$q(gXg^{-1}) = q(X), \qquad c(gXg^{-1}) = c(X)$$
for every $g \in PU(3)$. Moreover, for every $\lambda \in \mathbb{R}$,
$$q(\lambda X) = \lambda^2 q(X), \qquad c(\lambda X) = \lambda^3 c(X).$$
Hence
$$I(X) = \frac{c(X)^2}{q(X)^3}$$
is invariant under both the adjoint action and multiplication by nonzero scalars. Therefore I is a well-defined invariant of the induced action on $\mathbb{P}(V)$.

Now consider the family
$$X_a = i\operatorname{diag}(a, 1, -a-1), \qquad a \in \mathbb{R}.$$
A direct computation gives
$$q(X_a) = 2(a^2 + a + 1),$$
and
$$c(X_a) = -3a(a+1).$$
Therefore
$$I(X_a) = \frac{9a^2(a+1)^2}{8(a^2+a+1)^3}.$$
This is a nonconstant continuous function of a; for example,
$$I(X_0) = 0, \qquad I(X_1) = \frac{1}{6}.$$
Hence $I(X_a)$ assumes infinitely many values as a varies over $\mathbb{R}$.

Since I is constant on each $PU(3)$-orbit of $\mathbb{P}(V)$, distinct values of I correspond to distinct orbits. Therefore the action of $PU(3)$ on $\mathbb{P}(\mathbb{R}^8)$ has infinitely many orbits. $\square$

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